

Section - I: Multiple Choice Questions

1. What is a characteristic of a model that is underfitting the training data?
 - A) The model is too complex and performs well on the training data but poorly on new data.
 - B) The model is too simple and performs poorly on the training data as well as new data.
 - C) The model is just complex enough and performs well on both the training data and new data.
 - D) The model is too complex and performs poorly on the training data but well on new data.

Section - II: Descriptive / Case Study-Based Questions

1. A data analyst at a retail company is tasked with developing a machine learning model to predict customer churn based on historical data. The model is trained on a dataset containing 10 features, including customer demographics, purchase history, and loyalty program participation. After training the model, the analyst observes that the model performs extremely well on the training data, achieving an accuracy of 98%. However, when the model is tested on a separate validation set, the accuracy drops to 60%. What might be the problem with the model, and how would you address it?